RAG Learning

1.4 Powers, Roots and Standard Form

OCR GCSE Mathematics (J560) — Component 01 (Paper 1, Non-Calculator), Component 02 (Paper 2, Calculator), Component 03 (Paper 3, Calculator). H = Higher tier only.

1.4.1 Powers and Roots

- A **power** (or index/exponent) indicates repeated multiplication: $a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ times}}$
- a is the **base**; n is the **index/exponent/power**.
- The **square root** $\sqrt{a}$ is the positive number which, when squared, gives a: $\sqrt{a} \times \sqrt{a} = a$.
- The **cube root** $\sqrt[3]{a}$ is the number which, when cubed, gives a: $\sqrt[3]{a} \times \sqrt[3]{a} \times \sqrt[3]{a} = a$.

Square Numbers and Square Roots

n	n^2	$\sqrt{n^2}$	n	n^2
1	1	1	8	64
2	4	2	9	81
3	9	3	10	100
4	16	4	11	121
5	25	5	12	144
6	36	6	13	169
7	49	7	15	225

Cube Numbers and Cube Roots

n	n^3	$\sqrt[3]{n^3}$
1	1	1
2	8	2
3	27	3
4	64	4
5	125	5
10	1,000	10

Exam Tip: Every positive number has two square roots: a positive and a negative. For example $\sqrt{25} = 5$ or -5. However, the $\sqrt{}$ symbol denotes the **positive** root only. In exam questions, unless asked for "all solutions", give the positive root.

Worked Example:
Without a calculator, evaluate: (a) $\sqrt{196}$ (b) $\sqrt[3]{-27}$ (c) $(-2)^5$

(a) $14^2 = 196$, so $\sqrt{196} = 14$

(b) $(-3)^3 = -27$, so $\sqrt[3]{-27} = -3$

(c) $(-2)^5 = -2 \times -2 \times -2 \times -2 \times -2 = 4 \times 4 \times (-2) = 16 \times (-2) = -32$

1.4.2 Index Laws

The index laws allow you to simplify expressions involving powers of the same base.

Index Law Summary Table

Law	Rule	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$3^4 \times 3^2 = 3^6$
Division	$a^m \div a^n = a^{m-n}$	$5^7 \div 5^3 = 5^4$
Power of a Power	$(a^m)^n = a^{mn}$	$(2^3)^4 = 2^{12}$
Zero Index	$a^0 = 1$ (any non-zero a)	$7^0 = 1$, $\left(\frac{3}{4}\right)^0 = 1$
Negative Index H	$a^{-n} = \dfrac{1}{a^n}$	$2^{-3} = \dfrac{1}{8}$
Fractional Index H	$a^{1/n} = \sqrt[n]{a}$	$8^{1/3} = \sqrt[3]{8} = 2$
Fractional Index H	$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$	$8^{2/3} = (\sqrt[3]{8})^2 = 4$

Why does $a^0 = 1$?
Using the division law: $a^n \div a^n = a^{n-n} = a^0$. But any number divided by itself equals 1. Therefore $a^0 = 1$.

Worked Example:
Simplify: (a) $x^5 \times x^3$ (b) $y^8 \div y^2$ (c) $(z^3)^4$ (d) $2a^3 \times 5a^4$

(a) $x^5 \times x^3 = x^{5+3} = x^8$

(b) $y^8 \div y^2 = y^{8-2} = y^6$

(c) $(z^3)^4 = z^{3\times 4} = z^{12}$

(d) $2 \times 5 = 10$ and $a^3 \times a^4 = a^7$, so the answer is $10a^7$

Negative Indices H

Worked Example:

Evaluate: (a) 5^{-2} (b) $\left(\frac{2}{3}\right)^{-3}$ (c) $4^{-1/2}$

(a) $5^{-2} = \frac{1}{5^2} = \frac{1}{25}$

(b) $\left(\frac{2}{3}\right)^{-3} = \left(\frac{3}{2}\right)^{3} = \frac{27}{8} = 3\frac{3}{8}$

(c) $4^{-1/2} = \frac{1}{4^{1/2}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$

Fractional Indices H

The key insight: $a^{m/n} = \left(\sqrt[n]{a}\right)^m$.

When using fractional indices: always find the root first, then apply the power. This keeps numbers small and reduces errors.

Worked Example:

Evaluate: (a) $27^{2/3}$ (b) $16^{3/4}$ (c) $32^{-3/5}$

(a) $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$

(b) $16^{3/4} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$

(c) $32^{-3/5} = \frac{1}{32^{3/5}} = \frac{1}{\left(\sqrt[5]{32}\right)^3} = \frac{1}{2^3} = \frac{1}{8}$

Exam Tip: The index laws only apply when bases are the same. You cannot simplify $3^4 \times 2^3$ using the multiplication law — the bases differ.

1.4.3 Standard Form

Standard form (also called scientific notation) is a way of writing very large or very small numbers concisely:

$$A \times 10^n$$

where $1 \leq A < 10$ (A is between 1 and 10, not including 10) and n is an integer (positive or negative).

Writing Numbers in Standard Form

Move the decimal point until you have a number between 1 and 10. Count the number of places moved — this gives the value of n. Moving the point left means n is positive (large number); moving right means n is negative (small number).

Converting to standard form: move the decimal point until one non-zero digit remains before it.

Worked Example:

Write in standard form: (a) 350,000 (b) 0.00027 (c) 0.043

(a) $350{,}000 = 3.5 \times 10^5$ (decimal moves 5 places left)

(b) $0.00027 = 2.7 \times 10^{-4}$ (decimal moves 4 places right)

(c) $0.043 = 4.3 \times 10^{-2}$ (decimal moves 2 places right)

Worked Example:

Write as ordinary numbers: (a) 6.02×10^4 (b) 8.5×10^{-3}

(a) $6.02 \times 10^4 = 60{,}200$ (move decimal 4 places right)

(b) $8.5 \times 10^{-3} = 0.0085$ (move decimal 3 places left)

Calculations with Standard Form

On Paper 1 (non-calculator), manipulate the powers of 10 separately using index laws. On Papers 2 and 3, use the EXP or ×10ˣ button on your calculator and give answers in standard form where appropriate.

Worked Example:
Calculate $(3 \times 10^4) \times (2 \times 10^3)$.

$$3 \times 2 = 6 \quad \text{and} \quad 10^4 \times 10^3 = 10^7$$
$$= 6 \times 10^7$$

Worked Example:
Calculate $(8 \times 10^5) \div (4 \times 10^2)$.

$$8 \div 4 = 2 \quad \text{and} \quad 10^5 \div 10^2 = 10^3$$
$$= 2 \times 10^3$$

Worked Example:
Calculate $(5 \times 10^3) + (4 \times 10^2)$.

Convert to ordinary numbers first: $5000 + 400 = 5400 = 5.4 \times 10^3$

Alternatively: $5 \times 10^3 + 0.4 \times 10^3 = 5.4 \times 10^3$

Worked Example:
Calculate $(3 \times 10^4) \times (7 \times 10^6)$.

$$3 \times 7 = 21 \quad \text{and} \quad 10^4 \times 10^6 = 10^{10}$$

$$21 \times 10^{10} = 2.1 \times 10^{11}$$

(21 is not between 1 and 10, so adjust: $21 = 2.1 \times 10^1$, giving 2.1×10^{11})

Exam Tip: When multiplying standard form numbers, if your coefficient is not between 1 and 10, adjust it. For example, 21×10^{10} is not standard form — rewrite as 2.1×10^{11}.

Worked Example:
The distance from Earth to the Sun is 1.5×10^{11} m. Light travels at 3×10^8 m/s. How long does it take light to travel from the Sun to Earth? Give your answer in standard form.

$$\text{Time} = \frac{1.5 \times 10^{11}}{3 \times 10^8} = \frac{1.5}{3} \times 10^{11-8} = 0.5 \times 10^3 = 5 \times 10^2 \text{ seconds}$$

1.4.4 Surds H

Higher tier only. This section covers surds — an important topic for Higher GCSE students.

A **surd** is an irrational root — a square root (or other root) that cannot be simplified to a rational number. Examples: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, $\sqrt[3]{4}$. Non-examples (rational, not surds): $\sqrt{4} = 2$, $\sqrt{9} = 3$.

Surds give exact answers. Leaving an answer as $\sqrt{3}$ is more precise than writing $1.732...$

Simplifying Surds

To simplify $\sqrt{n}$, find the largest perfect square factor of n and take its square root outside the root sign.

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$$

Worked Example:
Simplify: (a) $\sqrt{48}$ (b) $\sqrt{75}$ (c) $\sqrt{200}$

(a) Largest square factor of 48: 16 $\sqrt{48} = \sqrt{16 \times 3} = \sqrt{16} \times \sqrt{3} = 4\sqrt{3}$

(b) Largest square factor of 75: 25 $\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$

(c) Largest square factor of 200: 100 $\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$

Adding and Subtracting Surds

Only **like surds** (same expression under the root) can be added or subtracted, just like collecting like terms in algebra.

Worked Example:
Simplify $3\sqrt{2} + 5\sqrt{2} - \sqrt{2}$

$$3\sqrt{2} + 5\sqrt{2} - \sqrt{2} = (3 + 5 - 1)\sqrt{2} = 7\sqrt{2}$$

Worked Example:
Simplify $\sqrt{12} + \sqrt{27}$

$\sqrt{12} = 2\sqrt{3};\ \sqrt{27} = 3\sqrt{3}$

$$2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$$

Worked Example:
Simplify $\sqrt{50} - \sqrt{8} + \sqrt{18}$

$\sqrt{50} = 5\sqrt{2};\ \sqrt{8} = 2\sqrt{2};\ \sqrt{18} = 3\sqrt{2}$

$$5\sqrt{2} - 2\sqrt{2} + 3\sqrt{2} = 6\sqrt{2}$$

Multiplying Surds

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\sqrt{a} \times \sqrt{a} = a$$

Worked Example:

Expand and simplify: (a) $\sqrt{3} \times \sqrt{12}$ (b) $(2+\sqrt{5})(3-\sqrt{5})$ (c) $(1+\sqrt{3})^2$

(a) $\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$

(b) Use FOIL:

$$(2+\sqrt{5})(3-\sqrt{5}) = 6 - 2\sqrt{5} + 3\sqrt{5} - (\sqrt{5})^2 = 6 + \sqrt{5} - 5 = 1 + \sqrt{5}$$

(c) $(1+\sqrt{3})^2 = 1 + 2\sqrt{3} + (\sqrt{3})^2 = 1 + 2\sqrt{3} + 3 = 4 + 2\sqrt{3}$

Rationalising the Denominator

It is standard practice to avoid surds in the denominator of a fraction. Rationalising removes the surd from the denominator.

Simple case: denominator is $\sqrt{a}$

Multiply numerator and denominator by $\sqrt{a}$.

$$\frac{p}{\sqrt{a}} = \frac{p \times \sqrt{a}}{\sqrt{a} \times \sqrt{a}} = \frac{p\sqrt{a}}{a}$$

Worked Example:

Rationalise the denominator of $\dfrac{6}{\sqrt{3}}$

$$\frac{6}{\sqrt{3}} = \frac{6 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

Worked Example:

Rationalise the denominator of $\dfrac{4}{\sqrt{8}}$

$$\frac{4}{\sqrt{8}} = \frac{4\sqrt{8}}{8} = \frac{\sqrt{8}}{2} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

Harder case: denominator is $a \pm \sqrt{b}$ H

Multiply numerator and denominator by the **conjugate** of the denominator: if the denominator is $a + \sqrt{b}$, multiply by $a - \sqrt{b}$ (and vice versa). This uses the difference of two squares: $(a+b)(a-b) = a^2 - b^2$.

Worked Example:

Rationalise the denominator of $\dfrac{5}{3+\sqrt{2}}$

Multiply by the conjugate $\dfrac{3-\sqrt{2}}{3-\sqrt{2}}$:

$$\frac{5}{3+\sqrt{2}} \times \frac{3-\sqrt{2}}{3-\sqrt{2}} = \frac{5(3-\sqrt{2})}{3^2-(\sqrt{2})^2} = \frac{15-5\sqrt{2}}{9-2} = \frac{15-5\sqrt{2}}{7}$$

Worked Example:

Write $\dfrac{1+\sqrt{5}}{2-\sqrt{5}}$ in the form $a + b\sqrt{5}$.

$$\frac{1+\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{(1+\sqrt{5})(2+\sqrt{5})}{4-5} = \frac{2+\sqrt{5}+2\sqrt{5}+5}{-1} = \frac{7+3\sqrt{5}}{-1} = -7-3\sqrt{5}$$

Answer: $-7-3\sqrt{5}$ (so $a = -7$, $b = -3$)

Exam Tip: Always simplify the surd before rationalising — it keeps the numbers smaller. For $\dfrac{4}{\sqrt{8}}$, note $\sqrt{8} = 2\sqrt{2}$, so the fraction becomes $\dfrac{4}{2\sqrt{2}} = \dfrac{2}{\sqrt{2}} = \dfrac{2\sqrt{2}}{2} = \sqrt{2}$.

Summary: Key Surd Rules

Rule	Formula	Example

Product	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{3} \times \sqrt{7} = \sqrt{21}$
Quotient	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$	$\frac{\sqrt{18}}{\sqrt{2}} = \sqrt{9} = 3$
Square of surd	$(\sqrt{a})^2 = a$	$(\sqrt{5})^2 = 5$
Simplify	$\sqrt{a^2 b} = a\sqrt{b}$	$\sqrt{72} = 6\sqrt{2}$
Rationalise (simple)	$\frac{p}{\sqrt{a}} = \frac{p\sqrt{a}}{a}$	$\frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$
Rationalise (conjugate)	$\frac{p}{a \pm \sqrt{b}} \times \frac{a \mp \sqrt{b}}{a \mp \sqrt{b}}$	$\frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$